

feature, which gives educators the power to share courseware materials between professors from the same institute or anywhere around the world. Professors can create lab descriptions and instructions, which can be made available on the oscilloscope.

These oscilloscopes are best-suited for research and development (R&D) applications, engineering measurements at educational institutions, and standard manufacturing and testing.

USB based and DIY oscilloscopes could be the right choice

Some industry experts are of the opinion that USB scopes have not yet reached the level of desktop models, but these are getting better with technological advancements in terms of measurement and reliability.

USB based scopes provide two important value additions to engineers—flexibility of converting a PC into a T&M platform and portability. In this era of the IoT, these scopes have a slightly greater edge over desktop versions. In fact, there are some interesting miniature devices that convert your laptops, tablets and mobile phones into a waveform analysing device at an affordable price. And, some of these are DIY type and open source, too.

Of course, not all USB scopes are as accurate as traditional oscilloscopes. For critical applications, USB devices would probably not be the right choice, but these are perfectly suited for educational institutions as well as hobbyists and engineers working on small- and mid-level applications on a tight budget.

With even major T&M companies (that are known for impeccable desktop oscilloscopes) continuing to add support to their range of USB oscilloscopes and introduce newer versions, it would not be wrong to predict that USB oscilloscopes in the future could completely replace desktop oscilloscopes in the basic

General Selection Criteria

Following specifications/features of an oscilloscope should be part of the selection criteria:

Bandwidth. Choose your bandwidth based on the edge speed (rise-time) of your signal. If you are unsure of your edge speed, you can use three times the clock rate for analogue signals or five times the clock rate for digital signals.

Application. Choose an oscilloscope according to the application like performing signal-integrity measurements or capturing intermittent signals.

Sample rate. For a Gaussian front-end, use four times the bandwidth. For a maximally flat (or brick wall) front-end, use 2.5 times the bandwidth.

Memory depth. Acquisition memory is the place where digitised samples are stored. Deeper memory allows you to keep your sample rate higher at slower time base settings, which allows you to maintain the full bandwidth longer. It is also useful when analysing waveforms particularly in single-shot mode. Look for a digital oscilloscope that remains responsive, even when deep memory is used.

Waveform update rate. When digital oscilloscopes are processing data, these cannot capture and display signals. Processing time is known as dead time. The shorter the dead time, the more likely you are to find infrequent events. In order to capture an infrequent signal, check if the digital oscilloscope slows down when you use features like digital channels, serial decode or more than the default memory. For debugging and troubleshooting, the higher the update rate, the better the oscilloscope.

Number of channels. The oscilloscope generally has an option to choose between two or four analogue channels and, additionally, have digital channels.

Probes. The oscilloscope probe is an integral part of the measurement system. These are classified as passive and active probes. Select an oscilloscope that provides connectivity to passive as well as active probes.

—Akash Srivastava, application engineer, Keysight Technologies India Pvt Ltd



For general-purpose measurement, prices of oscilloscopes vary from ₹ 12,000 to over ₹ 150,000. Therefore buyers need to be clear on their requirements for use, budget and expected lifetime of its use, while not forgetting the warranty that comes with it” — Abhishek Rao, technical marketing manager, element14 India



oscilloscopes segment for general-purpose usage, provided these are reliable and sturdy like traditional box-type instruments in their measurements and analysis. You can take a look at some of the interesting devices in this arena listed in Tables II and III.

Choose wisely

Between benchtop, USB and DIY, it could be a daunting task to choose the right oscilloscope for your needs. We have featured some of the hundreds of varying models with widely differing costs and specifications. It is important to invest some time considering the use case and application

area of your potential oscilloscope. It would be wise to follow the try-before-you-buy policy, read reviews and ask the T&M vendor to provide a demonstration.

Warranty is also an important factor to pay heed to. Tektronix provides a five-year warranty on their benchtop scopes, whereas other reputed companies such as Rigol, Keysight and Rohde & Schwarz offer three-year warranty on their range of desktop scopes. USB and DIY scopes typically come with a one-year warranty. Features such as bandwidth, sampling rate and memory depth are not upgradable in most of the oscilloscopes in the market. ●